QUOTIENT SHEAVES OF RELATIONS COMMUTE WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids-quotients.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The sheaf-theoretic shadow of the base change compatibility of quotient stacks.

Editorial note. The label lemma-base-change-quotient-sheaf names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter Lemma lemma-base-change-on-invariant (navigation stub, not delivered): for a pre-relation j on U with an R-invariant morphism to X, the base change along any X-prime to X gives U-prime equal to the fibre product of X-prime with U over X, R-prime equal to the fibre product with R, and an induced pre-relation j-prime over B
- Groupoids in Spaces, Section spaces-groupoids-section-quotient-sheaves: the quotient sheaf U/R is defined as the coequalizer of the two maps from R to U in the category of sheaves of sets on the fppf site
- Coequalizers in a comma category: a coequalizer diagram in the category of sheaves over the big site remains a coequalizer diagram after restricting to the comma category of sheaves over X
- Exactness of pullback in a topos: for any morphism of topoi the base change functor on sheaves preserves finite colimits, in particular coequalizers

Source reference. Original reference 048G: sources/stacks-groupoids-quotients/pinned-original-source.tex, source lines 298–298. Prerequisite pointer invoked at 354; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 044H: sources/stacks-chow/tags.txt, source lines 5400–5400. Prerequisite pointer invoked at 360; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In the situation of Lemma 048G there is an isomorphism of sheaves*

$$U'/R' = X' \times_X U/R$$

For the construction of quotient sheaves, see Groupoids in Spaces, Section 044H.

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Proof. Since $U \to X$ is R-invariant, it is clear that the map $U \to X$ factors through the quotient sheaf U/R. Recall that by definition

$$R \rightrightarrows U \longrightarrow U/R$$

is a coequalizer diagram in the category Sh of sheaves of sets on $(Sch/S)_{fppf}$. In fact, this is a coequalizer diagram in the comma category Sh/X. Since the base change functor $X' \times_X - : Sh/X \to Sh/X'$ is exact (true in any topos), we conclude. $\square$